Snowflake Architecture & Cache with Practical Examples

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1. Explain in your own words the three main layers of Snowflake architecture.

Practical:
Draw a simple diagram showing Storage, Compute (Virtual Warehouse), and Cloud Services.
Explain what happens when you run a SELECT query.

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2. What is the role of the Storage layer in Snowflake?

Practical:
Create a table and insert some data. Show how data remains stored even if the warehouse is suspended.

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3. Describe what a Virtual Warehouse is in Snowflake.

Practical:
Create a warehouse, run a query, then suspend the warehouse. Show how warehouse size affects query speed by resizing it.

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4. Explain micro-partitions and their importance.

Practical:
Run a query with a filter (e.g., WHERE AMOUNT > 1000) and explain how Snowflake reads only relevant micro-partitions to speed up the query.

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5. What is the Cloud Services layer?

Practical:
Show a login process and role-based access in Snowflake. Explain how Cloud Services verify permissions before query execution.

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6. Name and describe the three types of cache Snowflake uses.

Practical:
Run the same query twice and measure execution time to demonstrate Result Cache.
Restart warehouse and rerun query to explain Local Disk Cache.

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7. Explain Result Cache.

Practical:
Run a COUNT(*) query twice and note that the second run doesn’t consume compute resources.

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8. How does Local Disk Cache differ from Result Cache?

Practical:
Run a query, suspend warehouse, restart it, run the query again, and observe performance change.

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9. What happens to caches when table data changes?

Practical:
Run a query twice, insert new rows, run again and explain why the Result Cache is invalidated and query runs fresh.

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10. Describe the difference between first and second run of the same query.

Practical:
Run a query once (full compute), then immediately run again (Result Cache), and explain the difference in resource usage.

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11. How does Snowflake's compute scaling work?

Practical:
Run a complex query on a small warehouse, then increase warehouse size and rerun the query, observing improved speed.

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12. Explain external stage vs internal stage.

Practical:
Create an internal stage and upload a file. Create an external stage pointing to S3 and explain the differences.

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13. Explain the statement:

"Data never moves, compute comes and goes."
Practical:
Show data stored in table remains the same regardless of warehouse start/stop and resizing.

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14. Will Snowflake use Result Cache with non-deterministic functions like CURRENT_DATE? Why?

Practical:
Run a query with CURRENT_DATE, run it again, and show Result Cache is not used.

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15. Summarize how Snowflake optimizes cost through architecture and caching.

Practical:
Explain how suspending warehouses when idle and reusing Result Cache save money.